

# Fengzhe Zhang

Machine Learning Researcher • Probabilistic Generative Models

fengzhe.zhang.26@ucl.ac.uk | +44 (0)7421 727 978

## EDUCATION

### Gatsby Computational Neuroscience Unit, UCL

London, UK

PhD in Machine Learning and Computational Neuroscience

Oct 2026 – Sep 2030 (expected)

Research interests: generative models and causal inference.

Supervisor: Prof. Arthur Gretton. Fully funded by the Jump Trading Scholarship.

### University of Cambridge

Cambridge, UK

MPhil in Machine Learning and Machine Intelligence

Oct 2023 – Sep 2024

Distinction; final mark: 79.87%. Ranked 4th out of 36 students.

### Imperial College London

London, UK

BSc in Mathematics

Oct 2020 – Jun 2023

First Class Honours; final mark: 88.41%. Ranked 10th out of 251 students.

## WORK EXPERIENCE

### Member of Technical Staff Intern

Jump Trading

Mentors: Loren Puchalla Fiore, Léo Heidelberger

Jul 2026 – Sep 2026

Conducted research in deep learning and alpha modelling.

### Research Assistant

University of Cambridge

Supervisor: Prof. José Miguel Hernández-Lobato

Nov 2024 – Nov 2025

Developed and applied diffusion models and normalising flows to improve the efficiency and accuracy of molecular sampling.

## PUBLICATIONS & DISSERTATION

**Fengzhe Zhang**, Laurence I. Midgley, and José Miguel Hernández-Lobato. “[Efficient and Unbiased Sampling from Boltzmann Distributions via Variance-Tuned Diffusion Models](#).” *Accepted for publication in TMLR*, 2025. **J2C Certification; ICML 2026 poster** (Journal-to-Conference track).

**Fengzhe Zhang\***, Jiajun He\*, Laurence I. Midgley, Javier Antorán, and José Miguel Hernández-Lobato. “[Efficient and Unbiased Sampling of Boltzmann Distributions via Consistency Models](#).” *Machine Learning and the Physical Sciences Workshop, NeurIPS*, 2024. (\*Equal contribution.)

**Fengzhe Zhang**. “[Sampling of Molecular Energy Functions via Consistency Models](#).” *MPhil dissertation, University of Cambridge*, 2024.

## RESEARCH EXPERIENCE

### MPhil Dissertation

University of Cambridge

Supervisor: Prof. José Miguel Hernández-Lobato

Apr 2024 – Aug 2024

Developed a framework combining consistency models with importance sampling for molecular energy functions, achieving an approximately  $4\times$  sampling speedup over DDPM baselines; dissertation awarded Distinction.

### Undergraduate Research Assistant

Imperial College London

Supervisor: Dr Yingzhen Li

Jul 2023 – Oct 2023

Proposed and evaluated a method for directly controlling KL divergence during VAE training, eliminating  $\beta$ -parameter sweeps across multiple VAE architectures.

## TEACHING EXPERIENCE

### Peer Tutor and Teaching Assistant, Imperial College London

Oct 2022 – Apr 2023

Led first-year tutorials in Probability, Analysis, and Calculus; supported second-year Numerical Analysis teaching and Julia programming.

## SCHOLARSHIPS & AWARDS

**Jump Trading Scholarship** — sole recipient in the cohort

2026

**Dean’s List** — top 10% of students

2021, 2022, 2023

**Undergraduate Research Bursary** — awarded by the Department of Mathematics

2022

**Winton Prize** — outstanding second-year Mathematics group project

2022